



ADVANCED STATISTICS

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MODULE 4: DISCRETE PROBABILITY DISTRIBUTIONS

- ★ A discrete probability distribution function (PDF) has two characteristics:
1. Each probability is between zero and one, inclusive.
 2. The sum of the probabilities is one.

RANDOM VARIABLES AND THEIR PROBABILITY OF DATA

- **Random Variable (X)** - takes on a defined set of values with different probabilities
- Ex: If you roll a die, the outcome is random (not fixed) and there are 6 possible outcomes, each of which occur with probability one-sixth.
- Random variables can be discrete or continuous.
- **Discrete Random Variables** - have a countable number of outcomes
 - Ex: dead/alive, treatment/placebo, dice, counts, etc.
 - **Continuous Random Variables** - have an infinite continuum of possible values
 - Ex: blood pressure, weight, speed of a car, real numbers from 1 to 6

Example: A school counselor wants to know how many siblings students have. A random sample of **20 students** is surveyed. Let X be the number of siblings a randomly selected student has.

x	P(x)
0	3/20
1	7/20
2	6/20

3	3/20
4	1/20

- This is a **discrete probability distribution** because X takes the values 0, 1, 2, 3, 4 and:

$$\frac{3}{20} + \frac{7}{20} + \frac{6}{20} + \frac{3}{20} + \frac{1}{20} = 1$$

CUMULATIVE DISTRIBUTION FUNCTIONS (CDF)

- A probability function maps the possible values of x against their respective probabilities of occurrence, P(x)
- **P(x)** is a number from 0 to 1.0
- The area under a probability function is always 1.
- A CDF tells you the probability that X is **less than or equal to a certain value**.

Example: A teacher records the number of books read by 10 students during a month. Let X be the number of books a student read.

x	P(X=x)	F(x) = P(X≤x)
0	1/10	1/10
1	2/10	3/10
2	4/10	7/10
3	2/10	9/10
4	1/10	10/10 = 1

Example question: What is the probability that a student read **2 or fewer books**?

Solution: $F(2) = P(X \leq 2) = \frac{7}{10} = 0.7$

Answer: 0.7 or 70%

EXPECTED VALUES OF RANDOM VARIABLES

- **Expected Value** - often referred to as the “long-term” average or mean, denoted by the Greek letter μ

- This means that over the long term of doing an experiment over and over, you would expect this average.
- The **Law of Large Numbers** states that as the number of trials in a probability experiment increases, the difference between the theoretical probability of an event and the relative frequency approaches zero
- The theoretical probability and the relative frequency get closer and closer together
- **Expected Value** is just the weighted average or mean (μ) of random variable x
- Example: A student plays a game where they roll a die. If they roll a 6, they win ₱12. Otherwise, they win ₱0.
 - Solution: $P(6) = 1/6$
 $P(\text{not } 6) = 5/6$
 - Expected Value:
 $E(X) = (12)(1/6) + (0)(5/6)$
 $E(X) = 2$
 - Answer: The expected value is ₱2.

BINOMIAL DISTRIBUTION

- used if there are only TWO POSSIBLE OUTCOMES
- The outcomes that event (a) will occur and (b) will not occur.
- used when there are a fixed number of yes/no outcomes, such as win/lose, correct/wrong, success/failure
- Example: tossing a coin, rolling a die and wanting 6 as an outcome

BINOMIAL DISTRIBUTION FORMULA

$$P(x) = nCx p^x q^{n-x}$$

p = probability of success
 q = probability of failure ($1 - p$)
 n = number of experiments
 x = number being asked in the problem

Example: A student takes a 5-question true-or-false quiz. If the student guesses randomly, the probability of getting each question correct is 0.5. What is the probability that the student gets exactly 3 questions correct?

Given	$n = 5$ $p = 0.5$ $q = 0.5$ $x = 3$
Solution	$P(X = 3) = nCx p^x q^{n-x}$ $P(X = 3) = 5C3 0.5^3 0.5^{5-3}$ $P(X = 3) = 0.3125$
Final Answer	$P(X = 3) = 0.3125$ or 31.25%

- Easy way to identify Binomial Distribution: FITS
 - **F – Fixed number of trials**
 - The number of attempts is given.
 - Example: “A coin is tossed 10 times.”
 - **I – Independent trials**
 - The result of one trial does not affect the next.
 - Example: Each coin toss is independent.
 - **T – Two possible outcomes**
 - Each trial has only two outcomes: success/failure, yes/no, correct/wrong.
 - Example: A student answers a question correctly or incorrectly.
 - **S – Same probability**
 - The probability of success stays the same for every trial.
 - Example: Probability of getting heads is always 0.5.

Example: A basketball player takes 8 free throws. The probability of making each shot is 70%. What is the probability of making exactly 5 shots?

It is binomial because:

- **Fixed:** 8 shots
- **Independent:** Each shot is separate
- **Two outcomes:** Make or miss
- **Same probability:** 70% chance of making each shot

BIGGEST CLUE: fixed number of repeated trials + only two outcomes + the same probability = BINOMIAL

GEOMETRIC DISTRIBUTION

- used when you REPEAT TRIALS UNTIL YOU GET THE FIRST SUCCESS
- Key Points:

- There are 2 outcomes: success or failure.
- You keep repeating until the first success.
- The probability of success, p , stays the same for every trial.
- The number of trials can continue until success occurs.

GEOMETRIC DISTRIBUTION FORMULA

$$P(X = n) = p q^{n-1}$$

p = probability of success
 q = probability of failure ($1 - p$)
 n = number of trials

Example: A basketball player has a 60% chance of making a free throw. What is the probability that the player makes their first successful shot on the 4th try?

Given	$p = 0.60$ $q = 0.40$ $n = 4$
Solution	$P(X = 4) = p q^{n-1}$ $P(X = 4) = (0.60)(0.40)^{4-1}$ $P(X = 4) = 0.0384$
Final Answer	$P(X = 4) = 0.0384$ or 3.84%

➤ 3 Main Clues for Geometric Distribution:

1. You keep repeating the trial until you get the first success.
2. Each trial has only 2 outcomes: success or failure.
3. The probability of success, p , stays the same each time.

- Keywords:
- “first success”
 - “until they succeed”
 - “first time”
 - “on the 3rd/4th/5th try”

BIGGEST CLUE: trial until first success = GEOMETRIC

HYPERGEOMETRIC DISTRIBUTION

- takes sample from TWO GROUPS ONLY
- used when you select a sample from a population WITHOUT REPLACEMENT and want to find the probability of getting a certain number from a

specific group

HYPERGEOMETRIC DISTRIBUTION FORMULA

$$P(x) = \frac{(K C k)((N - K) C (n - k))}{N C n}$$

N = total number of items in the population
 n = number of items selected (sample size)
 K = total number of items in the group of interest (successes)
 k = number of items from the group of interest that you want to select

Example: A company has 30 employees, including 12 managers and 18 regular employees. A committee of 7 employees is chosen randomly without replacement. What is the probability that the committee contains exactly 3 managers?

Given	$N = 30$ $n = 7$ $K = 12$ $k = 3$
Solution	$P(3) = \frac{(K C k)((N - K) C (n - k))}{N C n}$ $P(3) = \frac{(12 C 3)((30 - 12) C (7 - 3))}{30 C 7}$ $P(3) = \frac{(12 C 3)(18 C 4)}{30 C 7}$ $P(3) = 0.3307$
Final Answer	$P(3) = 0.3307$ or 33.07%

➤ 3 Main Clues for Hypergeometric Distribution:

1. There is a fixed population
 - Example: 30 employees, 20 balls, 50 cards
2. You select a sample from that population
 - Example: Choose 5 employees or draw 4 balls
3. You sample **WITHOUT REPLACEMENT**
 - Once something is selected, it is not put back.
 - This means the probability changes after each selection.

BIGGEST CLUE: without replacement = HYPERGEOMETRIC

POISSON DISTRIBUTION

- used to find the probability of a certain number of events occurring within a FIXED INTERVAL OF TIME OR SPACE

- Key Idea: Count how many times an event happens in a given interval

POISSON DISTRIBUTION FORMULA

$$P(x) = \frac{\mu^x e^{-\mu}}{x!}$$

$e = \text{approx. } 2.71828$

$\mu = \text{mean number of occurrences (mean} = np)$
 $x = \text{number of occurrences/events you want to find the probability of}$

Example: A coffee shop receives an average of 4 customers per hour. What is the probability that exactly 6 customers arrive in one hour?

Given	$\mu = 4$ $x = 6$ $e = 2.71828$
Solution	$P(x) = \frac{4^6 (2.71828)^{-4}}{6!}$ $P(x) = 0.1042$
Final Answer	$P(x) = 0.1042 \text{ or } 10.42\%$

- 3 Main Clues for Poisson Distribution:
- You are counting events/occurrences**
 - Example: number of customers, calls, accidents, or mistakes.
 - There is a fixed interval**
 - It can be time, area, distance, volume, or space.
 - Example: per hour, per day, per page, or per kilometer.
 - You are given an average rate**
 - Example: “an average of 5 customers per hour.”

BIGGEST CLUE: “How many times does an event occur in a certain interval?” = POISSON

MODULE 5: CONTINUOUS PROBABILITY DISTRIBUTIONS

- ★ It describes the probability of a variable that can take any value within a range.
- ★ It involves measurable values, such as height, weight, temperature, or time.

CONTINUOUS RANDOM VARIABLES AND THEIR PROBABILITY DISTRIBUTION

- **Continuous Random Variable** - can take on an infinite number of values within a given range

CHARACTERISTICS OF CONTINUOUS RANDOM VARIABLES

- **Infinite Values** - continuous random variables can take on any value in a given interval
- **Probability Density Function (PDF)** - describes the likelihood of the variable falling within a particular range
- **Cumulative Distribution Function (CDF)** - gives the probability that the random variable is less than or equal to a certain value

NORMAL DISTRIBUTION

- also known as **Gaussian Distribution** or **bell curve**
- data is symmetrically distributed with no skew (skew = 0; nasa gitna)
- the mean determines the position of the curve
 - Increasing mean = right curve
 - Decreasing mean = left curve
- the standard deviation stretches or squeezes the curve
 - Small SD = narrow curve
 - Large SD = wide curve

STANDARD NORMAL DISTRIBUTION

- also called the **Z-Distribution**
- a special normal distribution where the mean is 0 and the standard deviation is 1
- Every normal distribution is a version of the standard normal distribution that's been stretched or squeezed and moved horizontally right or left.

Z-SCORE FORMULA

$$Z = \frac{x - \mu}{\sigma}$$

$x = \text{individual value}$
 $\mu = \text{mean}$
 $\sigma = \text{standard deviation}$

- The z-score could either be positive or negative.
- the corresponding value of the computed z-score in the z-table is its probability
- if you need a positive z and you are only provided with the negative z-table, use its negative counterpart in the table then subtract that value to 1 (and vice versa)

Example: The average height of students is 165 cm, with a standard deviation of 5 cm. What is the probability that a student is shorter than 160 cm?

Given	$\mu = 165$ $\sigma = 5$ $x = 160$
Solution	$Z = \frac{160-165}{5}$ $Z = -1$ $P(Z < -1) = 0.1587$
Final Answer	$P(Z < -1) = 0.1587 \text{ or } 15.87\%$

- 4 Things to Remember for Z-Score / Normal Distribution:
 1. A mean (μ)
 2. A standard deviation (σ)
 3. A specific value/score (x)
 4. It asks for a probability or percentage
- Keywords:
 - Probability
 - Percentage
 - Above / greater than
 - Below / less than
 - Between
 - Normally distributed
 - Mean and standard deviation

BIGGEST CLUE: Normal distribution + mean + standard deviation + probability = Z-SCORE

EXPONENTIAL DISTRIBUTION

- used to describe the WAITING TIME BETWEEN EVENTS occurring at a constant rate
- fixed time interval (waiting time until the next event)
- denoted by λ (lambda) = rate (parameter that defines the shape of the curve

EXPONENTIAL DISTRIBUTION FORMULAS	
$x = \text{specific waiting time}$ $\lambda = \text{rate of events per unit time}$ $e = \text{approx. } 2.71828$	
$f(x) = \lambda e^{-\lambda x}$	Probability Density Function (PDF): x is exact
$F(x) = P(X \leq x)$ $= 1 - e^{-\lambda x}$	Cumulative Distribution Function (CDF): less than or equal to x
$P(X \geq x) = e^{-\lambda x}$	more than or equal to x
$P(a < X < b) = F(b) - F(a)$	between a and b

Example 1 (PDF): A store receives an average of 2 customers per hour. Find the value of $f(x)$ when $x = 0.5$ hour.

Given	$\lambda = 2$ $x = 0.5$
Solution	$f(0.5) = 2e^{-(2)(0.5)}$ $f(0.5) = 0.7358$
Final Answer	$f(0.5) = 0.7358$

- Note: $f(x)$, a pdf, gives the density at a particular value of x , rather than the probability of exactly that value

Example 2 (CDF): A customer arrives at a store at an average rate of 2 customers per hour. What is the probability that the next customer arrives within 30 minutes?

Given	$\lambda = 2$ $x = 0.5$
Solution	$P(X \leq 0.5) = 1 - e^{-(2)(0.5)}$ $P(X \leq 0.5) = 0.6321$
Final Answer	$P(X \leq 0.5) = 0.6321 \text{ or } 63.21\%$

Example 3 (more than x): A customer arrives at a store at an average rate of 2 customers per hour. What is the probability that you will wait more than 1 hour for the next customer?

Given	$\lambda = 2$ $x = 1$
Solution	$P(X \geq 1) = e^{-(2)(1)}$ $P(X \geq 1) = 0.1353$
Final Answer	$P(X \geq 1) = 0.1353 \text{ or } 13.53\%$

Example 4 (in between): A customer arrives at a store at an average rate of 2 customers per hour. What is the probability that the

next customer arrives between 20 and 40 minutes?

Conversion	$20\ mins = \frac{20min}{60min} = \frac{1}{3}\ hour$ $40\ mins = \frac{40min}{60min} = \frac{2}{3}\ hour$
Given	$\lambda = 2$ $a = \frac{1}{3}$ $b = \frac{2}{3}$
Solution	$P(\frac{1}{3} < X < \frac{2}{3}) = F(\frac{2}{3}) - F(\frac{1}{3})$ $= (1 - e^{-(2)(2/3)}) - (1 - e^{-(2)(1/3)})$ $= (1 - e^{-4/3}) - (1 - e^{-2/3})$ $= 0.7364 - 0.4866$ $= 0.2498$
Final Answer	$P(\frac{1}{3} < X < \frac{2}{3}) = 0.2498\ or\ 24.98\%$

➤ 3 Main Clues for Exponential Distribution:

1. It asks about waiting time
- Example: Time until the next customer arrives, time until a phone call comes, time until a machine breaks
2. There is an average rate (λ)
- Example: “An average of 4 customers arrive per hour.”
3. It asks for a probability involving time
- Within a certain time
 - More than a certain time
 - Between two times

BIGGEST CLUE: “How long do I have to wait until the next event?” = EXPONENTIAL

- **POISSON VS EXPONENTIAL:**
- Poisson = HOW MANY?
 - Exponential = HOW LONG?

MODULE 6: JOINT PROBABILITY DISTRIBUTION

- ★ **Joint Probability** - probability of TWO EVENTS HAPPENING TOGETHER
- ★ commonly designated as Event A and B or Event X and Y
- ★ $p(A\ and\ B)$ or $p(A \cap B)$
- ★ **Joint Probability Distribution** - shows probability distribution for TWO OR MORE RANDOM VARIABLES

Example: In a class, 60% of students like basketball, and 40% of students like both basketball and volleyball. What is the

probability that a randomly selected student likes both basketball and volleyball?

Given	$P(\text{Basketball})=0.60$ $P(\text{Basketball} \cap \text{Volleyball})=0.40$
Answer	$P(\text{Basketball} \cap \text{Volleyball})=0.40$

★ Joint probability = probability that two events happen together

TWO DISCRETE RANDOM VARIABLES

- **Probability Mass Function** - function over the sample space of a discrete random variable X which gives the probability that X is equal to a certain value
- **Join Probability Mass Function** of the random variable x and y is:

- $P_{x,y}(x,y) = P(X=x, Y=y)$
- where: $0 \leq P_{x,y}(x,y) \leq 1$

Example: A survey is conducted to see whether customers purchase an item after seeing an online advertisement.

- X = 0, if a customer does not purchase the item
- X = 1, if a customer purchases the item
- Y = 0, if the customer does not click the advertisement
- Y = 1, if the customer clicks the advertisement

	Y = 0	Y = 1
X = 0	0.50	0.20
X = 1	0.10	0.20

Question: What is the probability that a customer purchases the item and clicks the advertisement?

Solution: $P(X = 1, Y = 1) = 0.20$

Answer: 0.20 or 20%

CONDITIONAL PROBABILITY DISTRIBUTION

- likelihood of an event occurring based on the occurrence of a previous outcome
- **Bayes’ Theorem** - mathematical formula used in calculating conditional probability

CONDITIONAL PROBABILITY FORMULA

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

P = probability
A = Event A
B = Event B

BAYES' THEOREM FORMULA
(rearranged/expanded conditional probability formula)

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Example 1 (Conditional Probability): A class has 20 students. Of these, 12 students like basketball, and 8 of the 12 basketball fans are boys. What is the probability that a student is a boy given that the student likes basketball?

Given	Students who like basketball = 12 Boys who like basketball = 8
Solution	$P(\text{Boy} \text{Bball}) = \frac{\text{Boys who like bball}}{\text{Students who like bball}}$ $P(\text{Boy} \text{Bball}) = \frac{8}{12}$ $P(\text{Boy} \text{Bball}) = 0.6667$
Answer	$P(\text{Boy} \text{Bball}) = 0.6667 \text{ or } 66.67\%$

➤ Note: In conditional probability, “given that” tells you what to use as the denominator.

Example 2 (Bayes’ Theorem): A school surveyed its students about their participation in sports and their habit of bringing a water bottle to school. The survey found that 30% of the students are athletes, while 70% are not athletes. Among the athletes, 80% bring a water bottle to school, while 20% of the non-athletes bring a water bottle. If a randomly selected student is known to bring a water bottle to school, what is the probability that the student is an athlete?

Given	$P(\text{Athlete}) = 0.30$ $P(\text{Water Bottle} \mid \text{Athlete}) = 0.80$ $P(\text{Water Bottle} \mid \text{Not Athlete}) = 0.20$
Solution	$P(W) = P(W A)P(A) + P(W A')P(A')$ $P(W) = (0.80)(0.30) + (0.20)(0.70)$ $P(\text{Water Bottle}) = 0.24 + 0.14 = 0.38$ $P(A \mid W) = \frac{P(W \mid A)P(A)}{P(W)}$ $P(A \mid W) = \frac{(0.80)(0.30)}{0.38}$ $P(A \mid W) = 0.6316$
Answer	$P(A \mid W) = 0.6316 \text{ or } 63.16\%$

➤ Note: Bayes' Theorem is usually used

when you are given an observation/evidence and want to find the probability of its possible cause.

MARGINAL PROBABILITY DISTRIBUTION

- also known as **Unconditional Probability**
- likelihood of an event occurring ignoring external events (remains constant)
- can be obtained by summing the joint probabilities over all possible values of the other variables

Example: A school surveyed 20 students about their favorite sport and favorite school subject. The results are shown below:

	Math	English	Total
Basketball	4	6	10
Volleyball	2	8	10
Total	6	14	20

Question 1: What is the marginal probability that a randomly selected student prefers Volleyball?

Solution:
$$P(\text{Volleyball}) = \frac{10}{20} = 0.50$$

Answer: 0.50 or 50%

Question 2: What is the marginal probability that a randomly selected student prefers English?

Solution:
$$P(\text{English}) = \frac{14}{20} = 0.70$$

Answer: 0.70 or 70%

PROBABILITY DISTRIBUTIONS CHEAT SHEET				
TOPIC	MAIN IDEA	KEY CLUE	FORMULA	REMEMBER
Binomial	Fixed number of trials	“Exactly ___ successes out of ___ trials”	$P(x) = nCx p^x q^{n-x}$	FIXED trials
Geometric	Trials until the first success	“First success on the ___ try”	$P(X = n) = p q^{n-1}$	FIRST success
Hypergeometric	Sampling without replacement	“Choose/select without replacement”	$P(x) = \frac{(K C k)((N-K) C (n-k))}{N C n}$	NO replacement
Poisson	Number of events in a fixed interval	“How many happen in 1 hour?”	$P(x) = \frac{\mu^x e^{-\mu}}{x!}$	HOW MANY?
Continuous	Variable can have any value in a range	Height, weight, time, temperature	Probability over an interval	MEASURABLE
Z-score / Normal	Find probability using mean & SD	“Above, below, or between”	$Z = \frac{x - \mu}{\sigma}$	STANDARDIZE
Exponential	Waiting time until the next event	“How long until the next?”	$f(x) = \lambda e^{-\lambda x}$ $F(x) = P(X \leq x) = 1 - e^{-\lambda x}$ $P(X \geq x) = e^{-\lambda x}$ $P(a < X < b) = F(b) - F(a)$	HOW LONG?
Joint Probability	Two events happening together	“A and B”	$P = A \cap B$	AND = BOTH
Conditional Probability	Probability given another event	“A given B”	$P(A B) = \frac{P(A \cap B)}{P(B)}$	GIVEN = LIMIT
Bayes’ Theorem	Find the probability of a cause using evidence	“Given that ___ happened, what is the chance of ___?”	$P(A B) = \frac{P(B A)P(A)}{P(B)}$	REVERSE the GIVEN
Marginal Probability	Probability of one variable regardless of the other	Row/column totals	Add the row or column, then divide by total	TOTALS